

AKHILESH KUMAR MISHRA

FULL STACK ENGINEER (MERN) | REACT · NODE.JS · MONGODB — PAYMENTS AT 2M+ USERS

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EXPERIENCE

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Held AEPS and MATM settlement at a **99% success rate** through peak load, by making the UPI and XML payment path idempotent — request keys, exponential-backoff retries and an automated end-of-day reconciliation job.
- Cut **p95 read latency 40%** on the transaction APIs serving **2 million+ users**, by collapsing N+1 **Mongoose** queries into a single **MongoDB** aggregation pipeline and compound-indexing the hot collections.
- Streamed **500,000+ daily transactions** to operations in real time, by building the transaction and geolocation dashboards in **React.js** and **Redux Toolkit** over a **WebSocket** feed with memoised selectors.
- Contained a bad release to one service instead of the whole portal for **2 million+ users**, by decomposing a monolithic e-governance system into **Node.js** and **Express.js** microservices behind a versioned REST gateway.
- Blocked **99.9%** of unauthorized access attempts on the payments API, by adding **JWT** access/refresh rotation, **Redis**-backed sessions, device binding, RBAC middleware and per-device OTP rate limiting.
- Retired manual releases across all **3** payment platforms, by containerising the services with **Docker** and gating every merge on **GitHub Actions** running **Jest** unit and **Supertest** integration suites.

Software Engineer (Contract)

Independent Client Engagement

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Shipped a field-workforce platform used by **1,000+ agents** end to end — schema design through AWS deployment — on **React.js**, **Node.js**, **Express.js** and **MongoDB**.
- Brought dashboard load time down **30%**, by route-level code splitting of the **React.js** map view, virtualising marker rendering and streaming positions over **WebSockets** instead of polling.
- Removed **45%** of redundant API calls, by moving server state onto **React Query** with stale-while-revalidate caching and consolidating client state in **Redux Toolkit**.

SELECTED PROJECTS

Himalayan Edges — MERN Commerce PWA

Lifted page-load speed and client usage 30% at a 20% cart conversion rate, by shipping a service-worker offline cache, route-level code splitting and S3-backed asset delivery on a React, Express and MongoDB storefront.

STACK: React.js / Node.js / Express.js / MongoDB / PWA / AWS S3

DigiPay Web SDK & React Widgets

Embeddable React component library and browser SDK published from an npm workspaces monorepo, giving merchant sites a drop-in payment UI over a versioned REST API. github.com/GitHub-akhilesh/Django_apis_digipay

STACK: React.js / TypeScript / npm workspaces / Webpack / REST

Task Platform — React Web and React Native

React web client and React Native mobile app over a containerised REST backend, built as a workspace monorepo with Webpack builds and Docker Compose orchestration. github.com/GitHub-akhilesh/To-do-list-app

STACK: React.js / React Native / Node.js / Docker / Webpack

Real-Time Geofencing & Telemetry Dashboard

Delivers live GPS positions and geofence alerts to 1,000+ field agents with no polling anywhere in the path, by streaming over WebSockets from an Express ingestion API into time-indexed MongoDB collections.

STACK: React.js / Node.js / Express.js / WebSockets / MongoDB / Redis

SUMMARY

500,000+ payment transactions a day for 2 million+ users on India's national e-governance platform. Cut p95 read latency 40% by rewriting the MongoDB aggregation pipelines behind it, and held AEPS and MATM settlement at a 99% success rate by making the payment path idempotent and self-reconciling. React, TypeScript and Node.js end to end — schema design through Dockerised CI/CD.

EDUCATION

Bachelor of Technology (B.Tech.)

Kanpur Institute of Technology

2019 – 2023 | Kanpur, UP

SKILLS

FRONTEND

React.js Next.js SSR/ISR TypeScript JavaScript (ES6+)

Redux Toolkit React Query Tailwind CSS Material UI

code splitting Core Web Vitals

BACKEND & APIS

Node.js Express.js REST API design OpenAPI/Swagger

WebSockets JWT OAuth 2.0 RBAC rate limiting

idempotency keys API versioning

DATA & CACHING

MongoDB aggregation pipelines compound indexing

sharding Mongoose Redis PostgreSQL

query profiling schema design

TESTING, CLOUD & PRACTICE

Jest React Testing Library Supertest Docker

GitHub Actions CI/CD AWS EC2/S3/CloudFront Nginx

Grafana Sentry Agile/Scrum code review

CERTIFICATIONS

IIT Kanpur — Cyber Security Certification
Advanced MERN Stack Developer Certification
Data Structures & Algorithms Certification
Database Systems & SQL Certification

ACHIEVEMENTS

- First Class Honors, B.Tech. — top academic rank in cohort.
- HackerRank 4-Star Gold badge in SQL.